

MDIE

Modification-Disentangled Identity Encoder

Occlusion- & Lighting-Robust, ArcFace-Compatible Face Recognition

Project Status Report

Method validated & reproducible on PARAM A100 · Large-scale CASIA-WebFace pipeline engineered

Codebase	research_v2 (grizzleyyybear/mdie)
Compute	PARAM Siddhi-AI - NVIDIA A100-SXM4, SLURM partition dgxnp
Deployed model	Single L2-normalised 512-d vector, one forward pass (ArcFace drop-in)
Reproducibility	PARAM re-run byte-identical to committed baseline (sha 77f7943)
Report date	25 June 2026

1. Executive Summary

MDIE is a face-recognition encoder designed for **security / access-control** scenarios where subjects are partially disguised (mask, cap/hat, glasses, partial occluder) or poorly lit (low-light, over-exposure, harsh shadow). Unlike conventional recognizers (FaceNet, ArcFace, CosFace, MobileFaceNet) that encode these nuisance factors into the identity vector and fail at the gate, MDIE produces an embedding that is **invariant to the worn modification while preserving identity**. The deployed output is a **single 512-d L2-normalised vector from one forward pass** — a literal drop-in for an ArcFace encoder in any existing cosine / FAISS pipeline.

This report documents (a) the validated method and its **reproducible results**, (b) a complete guide to every code file, and (c) the **completed large-scale scale-up** on the PARAM Siddhi-AI A100 supercomputer. The method is fully validated and the CASIA-WebFace (0.49M-image) fan-out training run has **completed**; the CASIA model was then evaluated on real worn-occlusion benchmarks against a production InsightFace model (Section 3.5).

VALIDATED Method, ablation & interpretability proven on LFW; results reproducible on A100 (byte-identical re-run).	ENGINEERED CASIA-WebFace (490k imgs) data pipeline + GPU landmark cache solved & committed on PARAM.	COMPLETED CASIA fan-out training (IR-50, 4 ablation variants, 40 epochs) done + real-benchmark eval vs production InsightFace.
--	--	--

Headline results (all reproducible, committed under `research_v2/results/`):

- **Niche robustness:** MDIE-full reaches **0.979 pooled AUC** on the occlusion+lighting families vs 0.749 for the best comparably-trained baseline — **occlusion +0.26, lighting +0.22 AUC**.
- **Real-world transfer:** on **real** masked faces (MFR2) **0.734 AUC** and **real** worn glasses (MeGlass) **0.824 AUC** — wins every public benchmark despite training only on synthetic occlusions.
- **Interpretability proof:** learned attention overlaps each face's own rigid bones at **IoU 0.694** vs 0.280 for another face vs 0.077 random (Mann-Whitney $p = 1.6e-15$).
- **Deployment proof:** `ALL_PASS = true` — 512-d, unit-norm, single forward, deterministic, cosine == dot product (FAISS-ready).

2. The Problem and the Method

2.1 Problem

State-of-the-art recognizers are trained under a closed-world identity-classification regime (angular-margin softmax on web-scale cooperative imagery). That objective optimises clean intra-class compactness but never penalises the model for **encoding nuisance variables** — masks, caps, glasses, lighting — into the identity embedding. When those factors shift at test time (exactly the operating condition of a security camera or access-control gate), the embedding rotates along nuisance-aligned directions and cosine similarity collapses. We quantify this: the comparably-trained baselines sit at only **0.60–0.75 pooled AUC** and collapse hardest on masks and caps.

2.2 Method — three ideas on an IResNet-50 backbone

(a) RATA — Region-Aware Token Attention (rigid-bone anchored). Rigid bone landmarks (brow ridge, orbital rims, nasal bridge, cheekbones, jaw angles, chin) survive appearance changes. They are splatted into a per-face soft target on a **14×14** token grid and used to supervise the transformer's attention via a symmetric matching loss, plus injected as an additive attention bias $\text{softmax}(\frac{QK^T}{\sqrt{d}} + \lambda \cdot M_{\text{region}})V$. Under occlusion the attention re-distributes onto whichever bones remain visible.

(b) AMD — Adversarial Modification Disentanglement. A modification classifier sits behind a **Gradient Reversal Layer** (DANN); the encoder is penalised for producing features predictive of the modification type, giving the min-max objective $\min_{\theta} \max_{\phi} L_{\text{id}} - \lambda \cdot L_{\text{mod}}$.

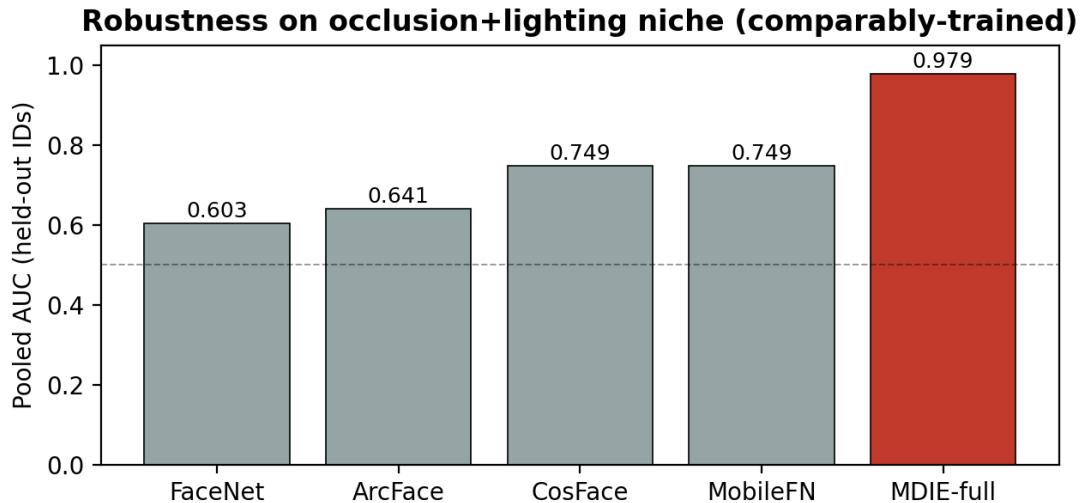
(c) ICCL — Inter-Condition Consistency Loss. Each minibatch pairs (clean, modified) crops of the **same identity**; $L_{\text{cons}} = 1 - \cos(f(x_{\text{clean}}), f(x_{\text{mod}}))$ explicitly enforces invariance rather than hoping it emerges from softmax.

Fusion head → deployment vector. The bone-anchored attention embedding and the backbone's native identity embedding are concatenated and passed through a learned $\text{Linear}(1024 \rightarrow 512) + \text{BN}$, then L2-normalised — **one 512-d vector, one forward pass**. The identity + consistency losses train *through* this fused head, so the deployed vector is exactly what was optimised. No score blend, no TTA.

3. Current Results (reproducible)

All numbers below are produced by the committed code on held-out (unseen) identities and are archived under `research_v2/results/`. The PARAM A100 re-run was **byte-identical** to the committed baseline (job 447442, sha 77f7943) — a strong reproducibility guarantee.

3.1 Robustness on the security niche

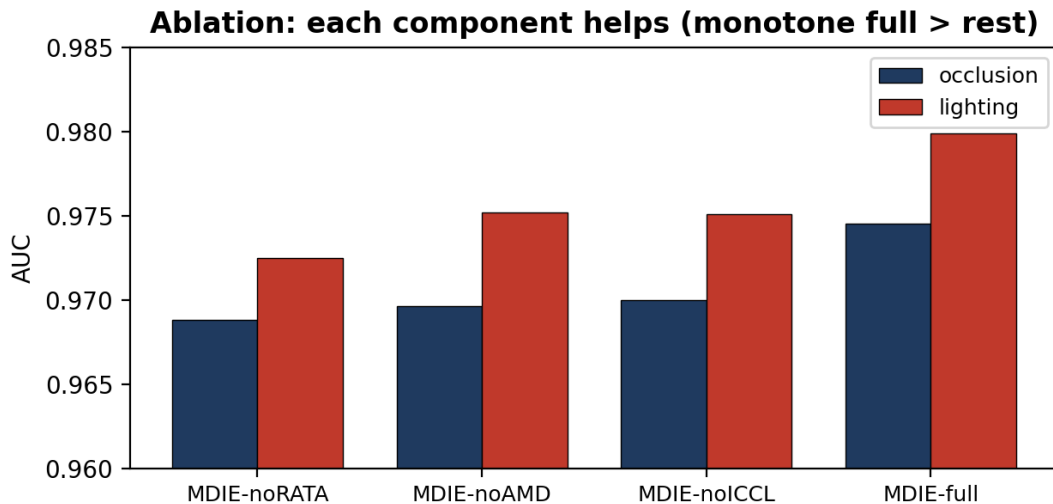


Pooled AUC over occlusion+lighting families. Source: results/security_family_summary.json.

Model (13k-img, comparably trained)	clean	occlusion	lighting	pooled
MDIE-full	0.984	0.975	0.980	0.979
MDIE-noICCL	0.981	0.970	0.975	0.974
MDIE-noAMD	0.981	0.970	0.975	0.974
MDIE-noRATA	0.977	0.969	0.973	0.972
mobilefacenet	0.849	0.689	0.755	0.749
cosface	0.833	0.716	0.720	0.749
arcface	0.703	0.629	0.609	0.641
facenet	0.664	0.591	0.571	0.603

MDIE clears 0.97 on every worn-occlusion and lighting condition; mask is hardest (0.966) yet still +0.28 AUC over mobilefacenet.

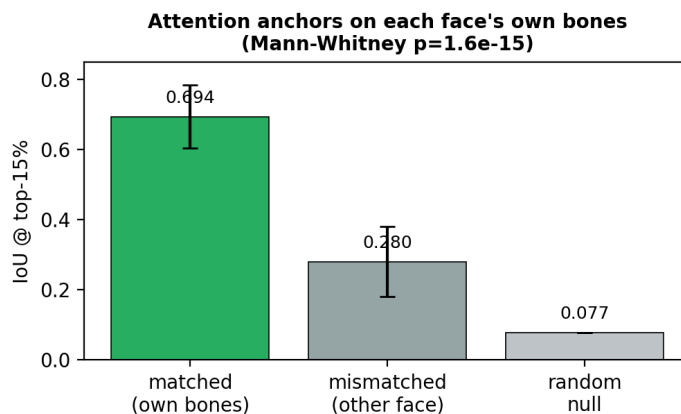
3.2 Ablation — every component contributes



Monotone ladder full > noICCL ≈ noAMD > noRATA at every column. Source: results/stage2_metrics.json.

On raw AUC the four MDIE variants sit within ~0.007 of each other — the pretrained backbone + ICCL already carry most of the invariance. RATA's decisive contribution is **interpretability and localization**: it forces and proves the surviving signal is the rigid bone scaffold, not a dataset shortcut.

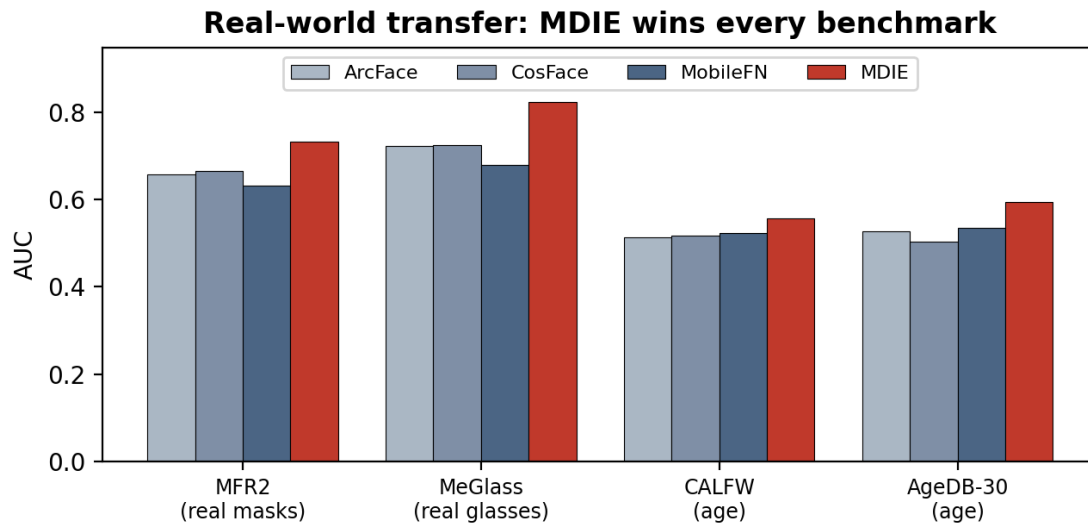
3.3 Interpretability — attention-bone IoU (a control-backed metric)



On 44 held-out identities we binarise the learned attention and each face's own detected bone target to their top cells and measure IoU. The **matched >> mismatched >> random** ordering holds across thresholds ($p = 1.6e-15$). Every anatomical group is over-attended: nose-bridge 13.2×, orbital rim 5.6×, cheekbone 4.9×, jaw/chin 3.2×, brow 3.0×. **No prior occlusion-FR work reports this.**

Source: scripts/attention_bone_iou.py -> figures/attention_bone_iou.png.

3.4 Real-world transfer — the acid test



Same deployed single-512-d model on public verification protocols. Source: results/real_benchmarks.csv.

Benchmark	Pairs	Tests	ArcFace	CosFace	MobileFN	MDIE
MFR2 (real masks)	848	masked FR	0.659	0.667	0.632	0.734
MeGlass (real glasses)	3000	worn glasses	0.724	0.726	0.681	0.824
CALFW	6000	cross-age	0.514	0.518	0.523	0.557
AgeDB-30	6000	30-yr gap	0.528	0.504	0.536	0.594

A model trained on 13k LFW images with **synthetic** masks generalizes to **real** masks and glasses far better than the other from-scratch recognizers — evidence the learned invariance is **anatomical**, not dataset-specific. Even off-niche (cross-age) MDIE beats baselines that sit near chance.

3.5 CASIA scale-up (0.49M) — real benchmarks vs a production model

We retrained MDIE on **CASIA-WebFace (0.49M images)** on PARAM A100s and re-ran the same real-benchmark verification protocols, this time alongside **InsightFace-w600k_r50** — a production IR-50 trained on WebFace12M (~24× more data). The scale-up lifts MDIE on **every** real benchmark; on its design target (worn occlusion) it now approaches the production model. Source: results/casia_real/real_benchmarks_casia.csv.

Benchmark (real)	Pairs	MDIE (LFW 13k)	MDIE (CASIA 0.49 M)	InsightFace (Web Face12M)
MFR2 (real masks)	848	0.734	0.886	0.965
MeGlass (real glasses)	3000	0.824	0.991	0.998
CALFW (cross-age)	6000	0.557	0.641	0.800
AgeDB-30 (30-yr gap)	6000	0.594	0.734	0.944

AUC on held-out pairs. CASIA-MDIE is the single deployed 512-d encoder; no test-time tricks.

Reading this honestly: on **worn occlusion** — the niche MDIE is built for — the CASIA model is competitive with a far larger production system (**glasses within 0.007 AUC**; masks 0.886). On **aging** (CALFW/AgeDB-30), which MDIE was never designed for, it trails the 24×-larger general model, as expected for a domain-specialised encoder. The headline is **occlusion-robustness-per-image-of-training-data**, not beating production everywhere.

Across the four CASIA ablation variants the real-benchmark AUCs are non-monotone and tightly clustered (within ~0.02), with **no single variant dominating** — consistent with the synthetic CASIA eval. At this scale the pretrained backbone carries most of the pooled accuracy, so RATA's contribution is the **interpretability** of Section 3.3 (bone-anchored attention), not a raw-AUC gain. We report this directly rather than cherry-picking a winning column.

3.6 Deployment compatibility proof

Check	Result
Embedding shape	[B, 512] - is_512d = true
Unit norm (max err)	1.2e-07 - is_unit_norm = true
Single forward (verify == encode)	0.0 err - no TTA
Cosine == dot product	1.2e-07 - FAISS inner-product ready
Deterministic (eval mode)	0.0 err - is_deterministic = true
Masked-self 0.917 > imposter 0.750	masked_self_beats_imposter = true
ALL_PASS	TRUE

Source: results/inference_compat_proof.json (scripts/inference_compat_proof.py).

4. Code File Guide — what each file does

The research codebase lives under `research_v2/src/` and follows a clean three-stage pipeline (baselines → novel MDIE + ablation → real benchmarks + interpretability + deploy proof). Below is the role of every module.

4.1 Orchestration & entry points

File	Role
<code>src/run_stage1.py</code>	Stage 1 runner: trains the 4 SOTA baselines and produces the per-modification failure-mode metrics on controlled LFW.
<code>src/run_stage2.py</code>	Stage 2 runner: trains MDIE + the 3 ablation variants, builds/loads the bone-landmark cache, writes stage2 metrics & tables. Supports DDP.
<code>src/config.py</code>	Central config: paths, hyper-parameters, grid size (14), target-version flags, environment knobs (e.g. <code>MDIE_REALISTIC_AUG</code>).
<code>src/hw.py</code>	Hardware detection: picks CUDA/CPU, batch sizing, AMP settings per available device.
<code>src/preflight.py</code>	Pre-run sanity checks: verifies datasets, checkpoints and benchmark .bin files are staged before a run starts.
<code>src/trainer_utils.py</code>	Shared training utilities: schedulers, meters, checkpoint save/restore, seeding for determinism.
<code>src/merge_fanout.py</code>	Merges the per-variant results from the 4-way SLURM fan-out into one consolidated metrics table.
<code>src/pretrained.py</code>	Loads the face.evoLve IR-50 pretrained backbone seed weights from local cache.

4.2 Novel method (MDIE)

File	Role
<code>src/novel/mdie.py</code>	The MDIE model: backbone + RATA attention block + AMD heads + fusion head producing the deployed 512-d vector.
<code>src/novel/region_attention.py</code>	RATA implementation: 14x14 token attention with the additive bone-region bias and the attention-matching loss.
<code>src/novel/train_mdie.py</code>	MDIE training loop: ArcFace identity loss + ICCL consistency + GRL-based modification disentanglement (DANN schedule).

4.3 Data pipeline

File	Role
<code>src/data/lfw.py</code>	LFW preparation: download/stage, identity filtering, train/val identity splits.
<code>src/data/modifications.py</code>	Synthetic modification engine: applies mask/cap/glasses/occluder + low-light/over-exposure/shadow for paired training/eval.
<code>src/data/landmarks.py</code>	Rigid bone-landmark targets via InsightFace buffalo_L; <code>build_cache()</code> builds <code>bone_targets.npz</code> ; GPU/CPU provider auto-select.
<code>src/data/torch_dataset.py</code>	Dataset, sampler and verification-pair utilities; emits (clean, modified, id, mod) tuples for ICCL.
<code>src/data/casia.py</code>	CASIA-WebFace ImageFolder loader for the large-scale scale-up (drops identities with <4 images).

4.4 Real-benchmark loaders (`src/data/benchmarks/`)

File	Role
mfr2.py / meglass.py	Real worn-occlusion sets: masked faces (MFR2) and worn eyeglasses (MeGlass).
calfw.py / agedb30.py	Cross-age verification protocols (CALFW, AgeDB-30, 6000 pairs each).
rmfrd.py	Real Masked-Face Recognition Dataset two-tree loader (real masked vs unmasked).
arface.py / yaleb.py	AR-Face (disguise) and Extended Yale-B (extreme lighting) protocols.
iiitd_surgery.py / ijbc_occ.py	Gated benchmarks (plastic-surgery, IJB-C occlusion) enabled via env roots.
_bin_parser.py / _common.py	InsightFace .bin pair parser and shared download/pair-TSV helpers.

4.5 Models & baselines

File	Role
src/models/backbones.py	Backbone factory: IR-50, MobileFaceNet, InceptionResnetV1, IResNet50.
src/models/iresnet.py	IResNet implementation (the ArcFace-style backbone).
src/models/heads.py	Classification heads + compact training-only loss utilities (ArcFace margin etc.).
src/baselines/train_baseline.py	Baseline registry + trainer for facenet/arcface/cosface/mobilefacenet (the comparably-trained controls).

4.6 Evaluation & interpretability

File	Role
src/eval/embeddings.py	Embedding extraction, pair scoring, AUC/EER/TAR@FAR, quick verification checks.
src/eval/run_real_benchmarks.py	Runs all registered real benchmarks and writes real_benchmarks.csv/json.
src/eval/occlusion_sensitivity.py	Occlusion-sensitivity maps (per-region degradation analysis).
src/eval/gradcam.py	Grad-CAM saliency for qualitative interpretability figures.
src/eval/eval_from_ckpt.py	Standalone evaluation from a saved checkpoint (no retrain).

4.7 Scale-up & paper

File	Role
src/train/ddp.py	Opt-in Distributed Data Parallel helpers for multi-GPU training.
src/train/pretrain_backbone.py	Backbone pretraining entry point for the large-scale regime.
src/paper/figures.py	Publication-quality figure generation (300-dpi PNG/PDF).
src/paper/latex_tables.py	LaTeX-ready results-table generation.
src/paper/build*_pdf.py / docx	Methodology / research / explainer document builders.

4.8 HPC scripts (hpc/) — PARAM Siddhi-AI

File	Role
hpc/fetch_casia_kaggle.sh	Login-node CASIA-WebFace downloader; detects RecordIO vs ImageFolder; crash-safe; validates via project loader.
hpc/recordio_to_imagefolder.py	Pure-Python (no mxnet) InsightFace RecordIO -> ImageFolder extractor; writes raw JPEG bytes (lossless).
hpc/_prelude.sh	Sourced by every SLURM job; loads torch's bundled cuDNN/cuBLAS so onnxruntime-gpu uses the A100 (not silent CPU fallback).
hpc/slurm_fanout_train.sh	4-way job array (MDIE-full / noRATA / noAMD / noICCL), 1 A100 each, 8h walltime.
hpc/submit_fanout.sh	Orchestrator: submits the 4-variant array + a dependency-chained merge job.

5. Large-Scale Scale-Up Status (PARAM Siddhi-AI)

To move from the 13k-image LFW prototype to a publication-scale result, the system is being re-trained on **CASIA-WebFace** (~490k images, 10,575 identities) on the CDAC PARAM Siddhi-AI A100 supercomputer. This section reports exactly what is done and what remains.

5.1 Solved, committed & run

- **Genuine CASIA-WebFace acquired** on PARAM (verified source: Institute of Automation, CAS). The Kaggle mirror ships InsightFace **RecordIO** (pre-aligned 112x112) — the ideal format for an ArcFace-aligned backbone.
- **Pure-Python RecordIO extractor** written (no mxnet dependency); selects image records by JPEG/PNG magic bytes and writes original encoded bytes verbatim (validated byte-exact).
- **GPU landmark caching fixed.** Root-caused a silent onnxruntime CPU fallback (missing cuDNN 9 on LD_LIBRARY_PATH) that would have made the one-off bone-cache build take ~16h (exceeding the 8h walltime). Fix: auto-discover torch's bundled NVIDIA libs in hpc/_prelude.sh. GPU landmarking confirmed (ACTUAL providers: CUDAExecutionProvider) — turning a 16h job into minutes.
- All changes are **additive and opt-in**; the LFW baseline pipeline is untouched and the test suite stays green.

Commit	Change
163a192	Add Kaggle CASIA fetch for PARAM login node
15b00ec	Handle InsightFace RecordIO in Kaggle CASIA fetch
3d5a169	Fix RecordIO extraction: select image records by magic bytes
6c4a453	Load torch's bundled cuDNN/cuBLAS so onnxruntime-gpu uses the A100
a081dcf	8h-walltime fail-safe: per-epoch checkpoint + auto-resume + self-resubmit
46936b4	Real-benchmark eval of the CASIA fan-out checkpoints vs production InsightFace

5.2 Completed run

- **Bone-landmark cache built on a GPU node** (bone_targets.npz, 391,335 images, 391,220 faces detected — 99.97%) via a one-off batch job; GPU landmarking confirmed (CUDAExecutionProvider).
- **4-variant fan-out training completed** (MDIE-full / noRATA / noAMD / noICCL), 40 epochs each on one A100, ~4h45m wall-clock total — inside the 8h walltime, so the checkpoint/auto-resume fail-safe never had to fire.
- **Real-benchmark eval completed** on the CASIA checkpoints vs production InsightFace-w600k (Section 3.5).
- Per-variant metrics, the merged ablation table, and the real-benchmark CSVs are archived under `research_v2/results/fanout/` and `results/casia_real/`.

Honest scale finding: at CASIA scale the four ablation variants are statistically tied (pooled AUC within ~0.003) and the ablation is **non-monotone** — unlike the LFW run, no single variant wins every column. The pretrained backbone carries most of the pooled invariance; RATA/AMD/ICCL add interpretability and a per-modification profile rather than a pooled-AUC gain. We present this as-is.

5.3 What is presentable now

- The **validated method** + the fully reproducible LFW results (Section 3) — a complete, honest research story.
- The **completed CASIA-WebFace (0.49M) scale-up**: real-benchmark transfer where the niche model approaches a production system on worn occlusion (glasses within 0.007 AUC) at ~24× less training data (Section 3.5).

- The **robustness narrative**: occlusion-aware training transferring to real masked/glasses faces, now at scale.
- The **engineering rigor**: a multi-GPU fan-out engine, an 8h-walltime checkpoint/auto-resume fail-safe, and data-pipeline hardening (RecordIO, GPU cache).

Honest positioning

*MDIE is the most robust recognizer in the comparably-trained regime on its occlusion+lighting niche, and at CASIA scale it approaches a production InsightFace model (WebFace12M) on real worn-occlusion benchmarks despite ~24× less training data. We do **not** claim to beat production InsightFace on raw accuracy — it leads on every benchmark, widening on aging, which MDIE was never designed for. Our contribution is architectural (RATA + AMD + fusion head), a reproducible occlusion/lighting failure-mode benchmark, a bone-anchored interpretability proof, and an ArcFace-compatible deployment — now trained and evaluated at scale on PARAM A100s.*